Flag relations and derived incidence geometries

Human Compression project

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Plan of the common theory

The two constructions studied here have different objects but share a practical route to their classification: describe chambers explicitly, identify every residue, and then check the remaining group action on a fiber. Part I retains the complete flag-relation theory for triangle geometries. Part II proves the corresponding residue and chamber statements for Steiner designs. Part III gives the complete unitary transport argument used by both applications.

The constructions and classification targets retain their original attribution. This edition assembles already accepted proofs, with their corrections; it asserts no new priority or additional source result. Each application supplies its own field, design and group hypotheses.

I. Flag relations and reconstruction

Scope and attribution

This paper supplies a reusable interface for triangle constructions on finite linear spaces. The construction is due to Delaby, Leemans and Tranchida [DLT, Construction 1]. The common-neighbor reconstruction, correlation extension and deleted-pencil description below reuse the project's earlier *Incidence compression*, §4. We give complete proofs so that the present paper is an actual dependency of an application, rather than a collection of references to an old combined manuscript. The point-fiber synchronization in Theorem 2 also covers two-point lines. The determinant criterion in Theorem 5 makes explicit the semilinear alternative omitted from DLT's affine-plane classification. Neither construction nor classical linear algebra is claimed as a new subject. The accompanying application identifies precisely which source statements these arguments recover or correct.

All sets in the geometric part are finite. A linear space S has a point set P, a line set L, at least two points on each line, at least two lines through each point, and a unique line through any two distinct points. A flag is an incident pair (p,L). A uniform space has k points on each line and r lines through each point; consequently v−1=r(k−1), where v=|P|. We use "automorphism" for a type-preserving incidence automorphism and "correlation" when interchange of points and lines is allowed. A duality need not have order two; a polarity does.

1. A directed relation on flags

Write F for the flag set. Define the relation D by

$$(p, L)D(q, M) \quad \Longleftrightarrow \quad p \in M \text{ and } q \notin L. \tag{1}$$

Because q belongs to M, the conditions imply p≠q and M=pq≠L. If A and B are the zero-one matrices for equality of point and equality of line on F, then

$$D = (A - I)(B - I). \tag{2}$$

Indeed the only possible intermediate flag is (p,M). This observation also proves that the integer product has entries zero or one.

The triangle incidence system Δ(S) has elements (f,i), where f∈F and i∈Z/3Z. Distinct types i,i+1 are incident when fDg, respectively; the reverse incidence is determined by symmetry. Its adjacency matrix, in cyclic type order, is

$$\begin{pmatrix} 0 & D & D^{\mathsf{T}} \\ D^{\mathsf{T}} & 0 & D \\ D & D^{\mathsf{T}} & 0 \end{pmatrix}. \tag{3}$$

The cyclic shift τ(f,i)=(f,i+1) is therefore a triality. If p0,p1,p2 are an ordered noncollinear triple, its chamber has flag of type i equal to (pi,p(i−1)pi), with subscripts read cyclically. Conversely, (1) forces these three joining lines in any chamber and excludes collinearity. A panel between (p,L,0) and (q,pq,1) has exactly one completion for each a∈L{p}: the third flag is (a,qa,2). Thus every flag extends to a chamber, and every panel has k−1 completions in the uniform case. In particular Δ(S) is firm precisely when k≥3 and is thin precisely when k=3.

For a general rank-two input, the existence of a chamber of the literal triangle system already forces a six-cycle in the input incidence graph: its three point labels and three line labels are distinct, consecutive incidences come from the chamber, and the required nonincidences prohibit repeated labels. Consequently, if that nonempty triangle system is a geometry, the input has gonality at most three. This recovers DLT Proposition 3.3; it does not assert that every input of such gonality gives a geometry.

The outgoing degree of the flag (p,L) is v−|L|, since each q outside L gives the unique successor (q,pq). Hence flag transitivity of Δ(S) forces every line of S to have the same size k: its automorphism group is transitive on each vertex type. The identity v−1=r_p(k−1), obtained by partitioning P{p} among the lines through p, then forces a constant r_p=r. This proves uniformity without assuming that the automorphisms of Δ(S) have already been reconstructed.

2. Reconstruction from common neighbors

Lemma 1 (point and line signatures). For distinct flags f,g in a uniform linear space, the following alternatives hold:

Relation between f and g	Common successors	Common predecessors
same point	(r−2)(k−1)	not needed
different points	0 or k−2	not needed
same line	not needed	(k−2)(r−1)
different lines	not needed	0 or r−2

For two flags (p,L),(p,M) at the same point, a common successor uses a line through p different from L and M, and then one of its k−1 other points. For flags at different points p,q, its line must be pq. If neither forbidden line is pq, its point may be any of the k−2 points on pq other than p and q; otherwise there are no choices. Reversing the reasoning proves the last two rows. Distinct lines either meet in one point or do not meet, so the reverse count has no further possibilities. This proves the lemma.

Theorem 2 (type-preserving reconstruction). If r≥3 and k≥2, restriction to the underlying flags identifies Aut(Δ(S)) with Aut(S).

The value (r−2)(k−1) is strictly larger than k−2. Thus the maximum off-diagonal common-successor relation on each type identifies precisely which flags have the same point. Every type-preserving automorphism φ of Δ therefore induces a permutation fi of P on type i. Write

$$\phi(p, L, i) = (f_i(p), M_i(p, L), i).$$

The point labels occurring among the successors of (p,L,i) are exactly P\L. Applying φ gives

$$P \setminus M_i(p, L) = f_{i+1}(P \setminus L), \qquad M_i(p, L) = f_{i+1}(L). \tag{4}$$

Since the image is a flag, fi(p) belongs to every f(i+1)(L) with L through p. At least two distinct lines pass through p and their intersection is {p}. Intersecting these images shows fi(p)=f(i+1)(p). All three point permutations are therefore a single permutation f, and (4) shows that f maps every line to a line and that φ is its diagonal action on flags. The inverse satisfies the same condition, so f is an automorphism of S. Conversely every automorphism of S preserves (1), proving the claimed equality. The same argument, with the target point set written P′, applies to any isomorphism Δ(S)→Δ(S′) between uniform spaces with the same parameters k,r: equation(4) becomes P′\M_i=f_(i+1)(P\L), and the same intersections synchronize the three point bijections. Thus every such triangle isomorphism is induced by one unique isomorphism S→S′. Notice that the proof never needs k−2>0; two-point lines are included.

If k,r≥3, both exceptional common-neighbor values in Lemma 1 are strict maxima. A correlation reversing the cyclic type order exchanges the successor and predecessor signatures and therefore exchanges the two flag partitions. On each type it therefore induces a bijection from points to lines and from lines to points. A point fiber and a line fiber meet in a flag exactly when their labels are incident, so these two bijections form a duality δi of S. In particular the action on type 0 is induced by one duality δ0. The globally defined lift of δ0 reverses the cyclic type order, since it transposes D. Composing the given correlation with its inverse lift gives a type-preserving automorphism which is the identity on type 0. Theorem 2 forces that automorphism to be the identity on every type. Thus all three actions were induced by δ0. No extra pointwise kernel has been suppressed in this reconstruction.

Theorem 3 (the complete correlation extension). For k,r≥3 there is a natural isomorphism

$$\mathrm{Cor}(\Delta(S)) \simeq C_3 \rtimes_\epsilon \mathrm{Cor}(S), \tag{5}$$

where ε records whether a correlation preserves or interchanges the two types of S, and ε=−1 acts on C3 by inversion. If S has no duality, (5) is C3×Aut(S). If S has a duality, the different exact sequence with kernel Aut(S) and quotient S3 splits if and only if S has a polarity.

For completeness, a correlation δ of S acts on a flag by δ(p,L)=(δp,δL) when it preserves types and by (δL,δp) when it interchanges them. Define

$$\widetilde{\delta}(f, i) = (\delta f, \epsilon(\delta) i).$$

This is a homomorphism into Cor(Δ), fixes type 0, and conjugates τ to τ raised to ε(δ). The reconstruction just proved shows that τ and all these lifts generate the full correlation group. The two displayed factors have trivial intersection, proving (5). A polarity supplies an involutory lift of a type transposition and, together with τ, a complement S3. Conversely any complement has such an involutory lift, whose induced duality of S has order two. This proves the splitting criterion. A polarity is not required for (5) itself; that qualification belongs only to the latter exact sequence.

The chamber correspondence and Theorem 2 give the useful equivalence

$$\Delta(S) \text{ is flag-transitive} \quad \Longleftrightarrow \quad \mathrm{Aut}(S) \text{ is transitive on ordered noncollinear triples.} \tag{6}$$

The same assertion holds for any specified subgroup of Aut(S), using its induced diagonal action.

3. Connectivity and the deleted-pencil residue

For k,r≥3, Δ(S) is connected. In a fixed type, flags with the same point have a common successor and flags with the same line have a common predecessor, by Lemma 1. These moves connect all flags because the incidence graph of a linear space is connected: two points have their joining line. Thus each type lies in one component, and an edge between successive types joins the three components.

Theorem 4 (residue interface). The residue at (p,L,0) is the bipartite incidence structure with parts

$$\{Q : Q \notin L\}, \qquad \{(a, M) : a \in L \setminus \{p\},\ M \neq L,\ a \in M\}, \tag{7}$$

and incidence Q∈M. For the first part, the corresponding original vertex is (Q,pQ,1); the second part already gives the original vertex of type 2. The residue in the Van Nypelseer construction used in L092 has precisely the same description with a allowed to range over all of L. Formula (7) is obtained by deleting one pencil, not by assuming that this deletion preserves connectivity.

To verify (7), incidence with (p,L,0) forces a type-1 vertex to be (Q,pQ,1) for Q outside L, and a type-2 vertex to be (a,M,2) with a∈L{p} and M≠L. Their mutual incidence requires Q∈M and a∉pQ. The latter condition is automatic, since a≠p and pQ meets L only in p. This proves both directions and the claimed literal identification.

In projective or affine dimension at least three, each edge in (7) stays inside the plane generated by L and Q. Different such planes have disjoint outside-point sets and disjoint allowed-line sets. Thus the residue is a disjoint union of its plane residues. Whenever those plane residues are connected, this is its exact component decomposition.

4. Determinants classify affine-plane frame orbits

We write vectors as rows. Let K be a finite field and H a subgroup between SL(d,K) and ΓL(d,K), with d≥2. A semilinear element (A,σ) acts as v↦v^σ A. For the affine group V⋊H, translating the first point of an ordered noncollinear triple to zero identifies its orbit with the H-orbit of an ordered independent pair of vectors.

Theorem 5 (frame-orbit criterion). If d≥3, H is transitive on ordered independent pairs. If d=2, its orbits on ordered bases are in bijection with the orbits on K* of its induced determinant-semilinear transformations

$$t \longmapsto \det(A)\,\sigma(t). \tag{8}$$

In dimension two, the affine group is transitive on ordered noncollinear triples exactly when the action (8) is transitive. For a linear H≤GL(2,K) this is equivalent to H=GL(2,K). For a semilinear H it need not imply containment of GL(2,K).

For d≥3, extend either ordered independent pair to a basis. A linear map sending one complete basis to the other may have determinant c≠1. Multiply one of the target basis vectors outside the given pair by c^{-1}. The resulting map has determinant one and still sends the required pair correctly.

For d=2, arrange an ordered basis as the two rows of an invertible matrix B. If B and C have equal determinant, there is exactly one matrix M=B⁻¹C with BM=C, and det M=1. Thus SL(2,K) acts regularly on every determinant fiber. A semilinear transformation sends B to B^σ A and its determinant to (8). These facts prove both implications of the orbit correspondence: an H-orbit has one orbit of determinant values; conversely first move a determinant to the desired value and then use the contained SL group within that fiber. The transformations (8) form a group because they are induced by an actual action on matrices. For a linear H, the determinant image is a subgroup of K* acting by multiplication. Transitivity forces the full image K*; together with its contained kernel SL(2,K), this gives H=GL(2,K).

The statements about equal-determinant fibers, semilinear determinant transport, and orbit saturation are formalized in `formal/DeterminantFibers.lean`. The geometric classification, finite-field subgroup identifications and connectivity proofs are written mathematics, not consequences already formalized in that file.

5. A classical projective-line normalizer interface

We record an elementary proof of the normalizer used by the circle branch of the application, so that it is not hidden in a higher-dimensional collineation theorem. This is a classical result, not a novelty claim.

Theorem 6. For the natural action on P¹(K), where K is any finite field,

$$N_{\mathrm{Sym}(\mathbf{P}^1(K))}(\mathrm{PSL}(2,K)) = \mathrm{P\Gamma L}(2,K). \tag{9}$$

Let f normalize PSL. Since PGL is sharply three-transitive and itself normalizes PSL, composing f with a projectivity lets us suppose that f fixes ∞,0,1. The stabilizer of ∞ in PSL consists of x↦ax+b with a a nonzero square (all nonzero a in even characteristic). Its translation subgroup U={x↦x+b} is its unique Sylow p-subgroup, since U is normal and the scalar factor has order prime to p. Thus f normalizes U. Because f fixes0, conjugation sends the translation by b to the translation by f(b), giving

$$f(x+b) = f(x) + f(b). \tag{10}$$

The stabilizer of ∞ and0 in PSL consists of the square multipliers. Since f also fixes1, conjugating the multiplier by a square s gives the multiplier by f(s), and hence f(sx)=f(s)f(x) for every nonzero square s; the assertion also holds for s=0. In even characteristic every field element is a square, so f is multiplicative. In odd characteristic every t∈K is a sum of two squares: the set of squares including0 and its translate t minus that set both have (q+1)/2 elements, so they intersect. Write t=s+u with s,u squares. Additivity gives

$$f(tx) = f(sx) + f(ux) = (f(s) + f(u))f(x) = f(t)f(x).$$

Thus f is a field automorphism in all cases. Undoing the initial projectivity proves containment in PΓL. Conversely projectivities and field automorphisms normalize PSL by their matrix action, proving equality. The proof includes q=2,3: the same translation and square arguments apply, without assuming simplicity of PSL. ■

References and dependency boundary

[DLT] R. Delaby, D. Leemans and P. Tranchida, *Geometries with trialities arising from linear spaces*, Algebraic Combinatorics 9 (2026), 379–402, DOI10.5802/alco.482. Construction (1) and the geometric targets are theirs.

[IC] *Incidence compression*, repository manuscript, §4. Its flag-matrix, correlation and deleted-pencil arguments are reused above. The application must charge this complete rewritten dependency; citing it does not make its proof cost disappear.

The group classification of locally two-transitive linear spaces remains an explicitly retained application input. The complete Hermitian norm-shell proof is included in Part III below. A source correction, an independently verified new question answer, and a recovered principal contract have different meanings and receive separate accounting.

II. Derived designs and chamber fibers

The design construction

Let $\mathcal{S} = S(t,k,v)$ be a finite Steiner system on a set Ω: every t-subset lies in exactly one block, and every block has k points. We assume $2 \le t \le k \le v$ and write

$$r = \frac{v-t+1}{k-t+1}.$$

This is the number of blocks through a fixed $(t-1)$-subset. Indeed, their complements to that subset partition the remaining points into sets of size $k-t+1$. Distinct blocks through that subset intersect precisely in it. These elementary observations will account for the exceptional incidences in the construction.

We use Construction 2 of Leemans [VN], with its attribution to Van Nypelseer and Meixner. The incidence structure $\Gamma(\mathcal{S})$ has ordinary objects of types $1, \dots, t-2$, namely the subsets of those sizes. Its three remaining types are marked objects

$$m = (T,E), \qquad |T| = t-1, \quad T \cap E = \emptyset, \quad T \cup E \text{ a block},$$

blocks B_0, and copied points $z \in \Omega$. Ordinary objects of different types are incident by inclusion. Their incidences with the last three types, and the incidences between those types, are

$$\begin{array}{llll} X \sim (T,E) \iff X \subseteq T, & & X \sim B_0 \iff X \subseteq B_0, & X \sim z \iff z \notin X, \\ (T,E) \sim B_0 \iff B_0 \cap (T \cup E) = T, & (T,E) \sim z \iff z \in E, & B_0 \sim z \iff z \notin B_0. & \end{array} \tag{S1}$$

Objects of the same type are incident only when equal. At $t = 2$ the ordinary part is empty. A marked object is then an original incident point–line pair, so (S1) is exactly the rank-three Construction 3.

Theorem S.1 (chambers and firmness). For a nontrivial Steiner system with $t \ge 2$, the construction is a firm geometry if and only if

$$k \ge t+1, \qquad r \ge 3. \tag{S2}$$

Under these conditions its panels missing an ordinary type have two completions. The panels missing the marked, block and copied-point types have respectively $k-t+2$, $r-1$ and $k-t+1$ completions.

Proof. The ordinary objects in a flag form a chain. Suppose first that a marked object (T,E) is present. Complete that chain inside T, choose an alternative block through T if none is present, and choose a copied point of E if necessary. Distinct blocks through T have disjoint tails, so these choices respect any already specified exceptional objects and give a chamber.

If a block B_0 is present but no marked object is specified, extend the largest ordinary set to a $(t-1)$-subset T of B_0. Use the prescribed copied point, or choose $z \notin B_0$ if none is prescribed. The latter is possible since $r \ge 3$ implies $v > k$. The unique block B' through $T \cup \{z\}$ differs from B_0 and meets it

in T. The marked object $(T, B' \setminus T)$ completes the required data. If neither a block nor a marked object is present, use or choose a copied point z outside the largest ordinary set, extend that set to $T \subseteq \Omega \setminus \{z\}$, and take the block on $T \cup \{z\}$ and any alternative block through T. Complete the chain inside T in each case. This also treats the empty ordinary chain when $t = 2$, and proves that every flag extends to a chamber.

For a missing ordinary type the bounding subsets differ by two elements, giving exactly two intermediate subsets. For a missing marked type let R be the largest ordinary set, of size $t-2$; put $R = \emptyset$ if $t = 2$. Choose $T = R \cup \{u\}$ with $u \in B_0 \setminus R$. The unique block through $T \cup \{z\}$ then determines the marked object, giving $k-t+2$ choices. The other two panel counts are the number of alternative blocks through T and the number of points of E. They are at least two under (S2).

Conversely, in a firm geometry a marked object belongs to a chamber. Removing the copied point or the alternative block gives panels with $k-t+1$ and $r-1$ completions. Both must have size at least two. If $r = 1$, no chamber contains that marked object in the first place. This proves necessity as well. □

In the remainder of this part we assume (S2).

Lemma S.2 (the induced automorphism group). Under (S2), the natural action identifies $\mathrm{Aut}(\Gamma(\mathcal{S}))$ with $\mathrm{Aut}(\mathcal{S})$, where automorphisms preserve every type.

Proof. An automorphism permutes the copied points. Their nonincidences with blocks recover exactly each block as a subset of Ω, so the induced permutation is a design automorphism. Nonincidences with an ordinary object recover its point subset too. For a marked object, its copied-point neighbors recover E. Its block neighbors are the $r-1 \geq 2$ alternative blocks through T; the intersection of any two of them is T. Thus T is recovered as well. All objects are therefore determined by this one point permutation. Conversely a design automorphism preserves all six rules (S1), proving the claim. □

Chambers as one stabilizer fiber

For a group acting equivariantly on a surjection $Y \to X$ between nonempty sets, transitivity on Y is equivalent to transitivity on X and transitivity of one point stabilizer on the corresponding fiber. To prove the converse, first move the image of a given element to the target image, and then move within that fiber. For the forward implication, an element carrying two points in the same fiber necessarily fixes their common image. We include this standard action argument to make its precise use clear.

Theorem S.3 (the chamber fiber). For every $G \leq \mathrm{Aut}(\mathcal{S})$ there is a G-equivariant bijection

$$\mathrm{Ch}(\Gamma(\mathcal{S})) \longleftrightarrow \left\{ ((x_1, \dots, x_t), B_0) : \begin{array}{l} x_1, \dots, x_t \text{ are distinct,} \\ \{x_1, \dots, x_{t-1}\} \subset B_0, \quad x_t \notin B_0 \end{array} \right\}. \tag{S3}$$

The fiber over each ordered t-tuple has size $r-1$. Consequently G is chamber-transitive if and only if it is t-transitive on Ω and the stabilizer of one ordered t-tuple is transitive on those $r-1$ blocks. In particular the chamber count is

$$(v)_t(r-1), \qquad (v)_t = v(v-1)\cdots(v-t+1). \tag{S4}$$

Proof. The full ordinary chain and its containing $(t-1)$-set T order the points of T: successive differences are singletons. Read these as $x_1, \dots, x_{t-1}$ and take x_t to be the copied point. The alternative block is B_0. Conversely the data on the right of (S3) determine the unique block B' on $T \cup \{x_t\}$. It differs from B_0, and the Steiner property gives $B' \cap B_0 = T$. The marked object is $(T, B' \setminus T)$ and the ordinary chain consists of the initial subsets of the tuple. The two constructions are inverse and commute with the group action. Of the r blocks through T, exactly one contains x_t. The preceding stabilizer argument and elementary counting now give all assertions. □

This also proves the exact formulation of Property $(*)$ in [VN, Theorem 9]. The source's Construction 1 has ordinary subsets of sizes $1, \dots, t-1$ and blocks, incident by inclusion. A chamber of our geometry is equivalently a chamber of Construction 1, an alternative block B' through its final $(t-1)$-set T, and a point $z \in B' \setminus T$. Forgetting z and then B' gives two surjective equivariant maps. Applying the stabilizer argument at each map says precisely that the action on Construction 1 is transitive, its chamber stabilizer is transitive on the alternative blocks, and the stabilizer of an alternative block is transitive on its remaining points. This is Property $(*)$. The alternative encoding (S3) proves that these conditions imply t-transitivity. Transitivity on the first choice alone would not suffice. The numbers of objects are

$$n_i = \binom{v}{i} \quad (1 \le i \le t-2), \qquad n_{\text{block}} = \frac{\binom{v}{t}}{\binom{k}{t}}, \qquad n_{\text{marked}} = n_{\text{block}} \binom{k}{t-1}, \qquad n_{\text{copy}} = v. \tag{S5}$$

The marked count chooses its block and then T. These formulas and the panel counts determine numerical labels, but do not identify a stabilizer group up to isomorphism.

Derivation and all residues

For $X \subseteq \Omega$ of size $j < t$, write $\mathcal{S}_X$ for the derived system: its points are $\Omega \setminus X$, and its blocks are $B \setminus X$ for blocks containing X. Its parameters are $S(t-j, k-j, v-j)$. In particular $k-t+1$ and r are unchanged. Let $\mathcal{B}(X)$ be the geometry of the nonempty proper subsets of a finite set X, with inclusion incidence and one type per size. A join makes every object of one factor incident with every object of another. Empty rank-zero factors are allowed.

Lemma S.4 (Boolean residues and the derived residue). The geometry $\mathcal{B}(X)$ and its truncation to all subset sizes $1, \dots, h < |X|$ are residually connected. Under (S2), the residue at an ordinary object X of size j is naturally

$$\Gamma(\mathcal{S})_X \simeq \mathcal{B}(X) * \Gamma(\mathcal{S}_X). \tag{S6}$$

Proof. A flag in a subset geometry is a chain. Its residue is the join of subset geometries on successive set differences, with a final truncated factor above its largest member. In a factor of rank at least two, singleton objects are connected through two-element objects, and every other object has a singleton neighbor. If two positive-rank factors remain, their complete cross-incidence connects the whole residue even when each factor has rank one. Rank-zero and rank-one residues impose no connectivity condition. This proves the first assertion and also proves that joins of residually connected geometries are residually connected.

Objects below X give $\mathcal{B}(X)$. Above X, send an ordinary Y to $Y \setminus X$, a marked (T, E) to $(T \setminus X, E)$, a block B_0 to $B_0 \setminus X$, and an incident copied point $z \notin X$ to itself. These are bijections onto the objects of $\Gamma(\mathcal{S}_X)$. Subtracting X preserves inclusions and all the exceptional rules (S1). Every object below X is incident with every object of this second factor, including its copied points. Thus (S6) is an incidence isomorphism, not just an equality of counts. □

Theorem S.5 (reduction to rank three). Under (S2), $\Gamma(\mathcal{S})$ is residually connected if and only if $\Gamma(\mathcal{S}_R)$ is residually connected for every $(t-2)$-subset R. At $t = 2$ this means the identity assertion with $R = \emptyset$. No transitivity assumption is required.

Proof. For necessity take a complete ordinary chain with largest member R, using every size from 1 to $t-2$. Its residue is exactly $\Gamma(\mathcal{S}_R)$: the complete chain removes the Boolean factors in (S6). Every flag residue of a residually connected geometry is residually connected. Connectedness of the join at R alone would not prove this implication, so the complete chain matters.

For sufficiency induct on t. The starting case $t = 2$ is tautological. For an ordinary X of size j, the final two-design derivatives of $\mathcal{S}_X$ are precisely the $\mathcal{S}_R$ with $R \supseteq X$, $|R| = t-2$. Induction and Lemma

S.4 imply that the residue at X is residually connected. This deals with every flag containing an ordinary object, including flags with additional exceptional objects.

It remains to inspect the subsets of the three exceptional types. At a marked object (T, E) the residue is the join of $\mathcal{B}(T)$, the $r-1$ alternative blocks through T, and the discrete set E of copied points. All cross-incidences are present. Fixing an additional block or copied point removes its discrete factor, so Lemma S.4 treats all four cases containing the marked object whenever the remaining rank is at least two.

At an incident block–copied-point pair (B_0, z), the marked objects are in bijection with the $(t-1)$-subsets T of B_0: use the unique block through $T \cup \{z\}$. The remaining incidence is precisely the subset geometry on B_0 with sizes $1, \dots, t-1$, which is connected through points and two-subsets for $t \geq 3$.

At a block B_0 alone, its ordinary singleton points and the copied points outside it form a nonempty complete bipartite graph. Every other ordinary or marked object has a singleton neighbor in B_0. At a copied point z alone, the ordinary and marked objects give all subsets of $\Omega \setminus \{z\}$ of sizes $1, \dots, t-1$; the largest subsets again determine their unique blocks through z. This connected inclusion geometry has a point neighbor on every remaining block. Finally the full graph contains the graph between ordinary singletons and copied points with adjacency $x \neq z$. It is $K_{v,v}$ minus its matching, connected since $v \geq 3$; every other object has a singleton neighbor. These are the remaining four cases. Together they exhaust the exceptional-only flags and complete the induction. □

Theorem S.6 (rank-two intersections). Under (S2), every rank-two residue of $\Gamma(\mathcal{S})$ is a partial linear space or a generalized digon. Thus the source's property $(\mathrm{IP})_2$ holds throughout this family.

Proof. At $t = 2$, the residue at an original flag (x, M) is complete bipartite between $M \setminus \{x\}$ and the other lines through x. At an original point p, a remaining flag is determined by its point x, since its line is px; incidence with the lines avoiding p is $x \in L$. This is a subincidence structure of the original linear space. At an original line L, a remaining flag (x, M) is determined by M, since $x = M \cap L$; its incidence with points outside L is again a subincidence structure of the original linear space. The latter two residues are partial linear spaces.

For higher t, a defining flag containing an ordinary object is handled by (S6). Residues of a join are joins of factor residues. If two positive-rank factors remain they both have rank one and give a generalized digon. If only the derived factor remains use induction. In a Boolean factor, a rank-two residue is either a join of two rank-one intervals or the incidence of singletons and two-subsets of a three-element set. These give the same two possibilities.

A defining flag containing only exceptional objects has size $t-1$, so here $t \leq 4$. For $t = 4$ all three exceptional objects are fixed and the remaining objects are the nonempty proper subsets of their three-set T, giving a triangle. For $t = 3$, a marked object and a block leave the two points of T and the copied points of E with complete incidence. A marked object and a copied point similarly leave the two points of T and the alternative blocks. A block B_0 and copied point z leave the points and two-subsets of B_0; each two-subset determines its marked object through z. This is a partial linear space. These cases exhaust all possibilities. □

The Boolean argument uses the complete-chain interval decomposition. It does not assert that arbitrary two-type truncations of a Boolean lattice are partial linear spaces. The source constructions, firmness criterion and transitivity target are retained from [VN]; the residue and fiber proofs above reuse the accepted phase182/186 proof and its independent phase187/199 reviews. The all-rank induction and its applications are written mathematics, not a claimed full Lean theorem.

Reference. [VN] D. Leemans, *A generalization of a construction due to Van Nypelseer*, Beiträge zur Algebra und Geometrie 46 (2005), 561–574, Constructions 2–3 and Theorems 1 and 9–11. The two re-

tained classification inputs used in the companion application are stated there with their exact hypotheses; they do not follow from S.5.

III. Hermitian transport and residues

Transport on a nonzero Hermitian shell

The difficult residues in both applications can be described on the same set of vectors. We give the complete algebraic argument here, including the maps that turn its group generators into actual incidence paths. These are elementary unitary calculations, not a claim of a new classification theorem. The proof combines the accepted unital argument with its explicit transport supplement.

Let K be a commutative field and σ an involutive field automorphism, written $a \mapsto \bar{a}$. On column vectors in K^2 put

$$h((x,y),(u,v)) = x\bar{v} + y\bar{u}, \qquad N(x,y) = x\bar{y} + \bar{x}y, \qquad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Write $\mathrm{SU}(h)$ for the determinant-one matrices preserving this form. For $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with determinant one, the equation $M^{\mathsf{T}} J \sigma(M) = J$ is equivalent to

$$\bar{a} = a, \quad \bar{d} = d, \quad \bar{b} = -b, \quad \bar{c} = -c, \qquad ad - bc = 1. \tag{H1}$$

Indeed solving for $\sigma(M)$ gives $\sigma(M) = J(M^{\mathsf{T}})^{-1}J = \left(\begin{smallmatrix} a & -b \\ -c & d \end{smallmatrix}\right)$; the calculation is reversible. Direct expansion also verifies $N(Mv) = N(v)$ under (H1).

Theorem H.1 (unique unitary transport). If $N(v) = N(w) = k \neq 0$, there is exactly one element of $\mathrm{SU}(h)$ carrying v to w. For $v = (x,y)$ and $w = (u,r)$ it is

$$T_{w,v} = \frac{1}{k}\begin{pmatrix} u\bar{y} + \bar{u}y & u\bar{x} - \bar{u}x \\ r\bar{y} - \bar{r}y & r\bar{x} + \bar{r}x \end{pmatrix}. \tag{H2}$$

Proof. The columns of

$$B_v = \begin{pmatrix} x & -\bar{x} \\ y & \bar{y} \end{pmatrix}$$

are orthogonal and have respective norms $k, -k$. Its determinant is k, so it is invertible. Thus $B_w B_v^{-1}$ has determinant one, preserves the Hermitian form, and carries the first column of B_v to that of B_w. Matrix multiplication gives (H2). The argument includes characteristic two: the two columns remain independent because their determinant is the nonzero element k.

For uniqueness, (H1) implies that any such matrix taking (x,y) to (u,r) must take $(-\bar{x},\bar{y})$ to $(-\bar{u},\bar{r})$. Its values on the basis formed by the columns of B_v are determined, so it is $B_w B_v^{-1}$. □

Two isotropic directions suffice

For $\bar{t} = -t$, the matrices

$$U_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad V_t = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix} \tag{H3}$$

satisfy (H1). Their displacements lie on the two isotropic coordinate axes. If $N(v) \neq 0$ and $t \neq 0$, those displacements are nonzero: neither coordinate of v can be zero.

Now let $K = \mathbb{F}_{q^2}$ with $\sigma(a) = a^q$. The trace-zero space has dimension one over $\mathbb{F}_q$. Choose a nonzero δ with $\bar{\delta} = -\delta$, including in characteristic two. Then $\delta^2 \in \mathbb{F}_q^*$, and the entries of (H1) satisfy $a, d \in \mathbb{F}_q$, $b, c \in \delta\mathbb{F}_q$. Conjugation by $\mathrm{diag}(\delta, 1)$ identifies these matrices with $\mathrm{SL}(2, q)$: explicitly, conjugating by its inverse on the left changes the off-diagonal entries to b/δ and $c\delta$, which range over $\mathbb{F}_q$. The two families (H3) therefore become all upper and lower elementary matrices over $\mathbb{F}_q$.

For clarity, these elementary matrices generate $\mathrm{SL}(2, F)$ over every field F. If $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ has $a \neq 0$, left multiplication by $V_{-c/a}$ and right multiplication by $U_{-b/a}$ give $\mathrm{diag}(a, a^{-1})$. If $a = 0$, then $c \neq 0$ and $U_1 A$ has nonzero upper-left entry. Finally, for $t \neq 0$,

$$W(t) = U_t V_{-t^{-1}} U_t = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}, \qquad W(t)W(-1) = \mathrm{diag}(t, t^{-1}). \tag{H4}$$

This proves generation, including over $\mathbb{F}_2$.

Theorem H.2 (connectivity using specified directions). Let W be a nondegenerate Hermitian plane over $\mathbb{F}_{q^2}$, let $k \in \mathbb{F}_q^*$, and let $\mathcal{A}$ be a set of its isotropic one-spaces containing at least two distinct members. The graph on $X_k = \{v \in W : h(v, v) = k\}$ which joins distinct v, w whenever $K(v - w) \in \mathcal{A}$ is connected. The group $\mathrm{SU}(W)$ acts regularly on X_k, and $|X_k| = q(q^2 - 1)$.

Proof. Choose isotropic representatives e, f for two specified directions. Their pairing is nonzero, since the orthogonal complement of an isotropic line in a nondegenerate plane is that same line. Rescale them to give Gram matrix J. Theorem H.1 proves regularity of the special unitary action on each nonempty nonzero shell. Such shells are nonempty: with $y = 1$ the equation $N(x, 1) = k$ is the surjective trace equation. The two root families in this chosen basis generate the whole group by (H4), and each nonidentity root element makes a nonzero move in one of the two allowed directions. An element carrying one shell vector to another can therefore be written as a product of moves along graph edges. This proves connectivity. Finally $|\mathrm{SL}(2, q)| = q(q^2 - 1)$, by choosing a nonzero first column and then one of the q second columns with determinant one. Regularity gives the claimed shell size. □

The projective incidence map

Let V be a nondegenerate Hermitian three-space over $K = \mathbb{F}_{q^2}$. Its classical unital has as points the isotropic one-spaces, and as blocks their intersections with secant projective lines. It is the classical $S(2, q + 1, q^3 + 1)$ design, with q^2 blocks through a point. These design parameters and elementary finite-field Hermitian linear algebra are the classical inputs used here.

Fix a block L. Its projective line corresponds to a nondegenerate two-space W. A degenerate two-space would have a one-dimensional radical and just one isotropic projective point, whereas a secant has $q + 1$. Write $V = W \perp Kz$ and use surjectivity of the field norm to arrange $h(z, z) = 1$. Every unital point outside L has a unique representative $v + z$, where

$$v \in W, \qquad h(v, v) = -1. \tag{H5}$$

For distinct such vectors v, w, the projective line joining $\langle v + z \rangle$ and $\langle w + z \rangle$ meets W in $\langle v - w \rangle$. Hence their joining block meets L at an isotropic point precisely when $v - w$ is isotropic. The point of intersection specifies its direction, with no arbitrary choice of a group isomorphism.

Corollary H.3 (full and deleted pencils). Let $\mathcal{A}$ be a set of at least two points on the unital block L. Consider the bipartite geometry whose first part is the unital points outside L, whose second part consists of flags (a, M) with $a \in \mathcal{A}$, $M \neq L$, and whose incidence is $Q \in M$. This graph is connected for every prime power $q \geq 2$.

Proof. By (H5), its point-collinearity graph is exactly the shell graph of Theorem H.2 for the directions represented by $\mathcal{A}$. Every flag vertex has q outside-point neighbors, so connectivity of the point-collinearity graph connects the entire bipartite graph. □

Taking $\mathcal{A} = L$ gives the block residue of the Van Nypelseer construction. Taking $\mathcal{A} = L \setminus \{p\}$ gives the triangle residue in Part I, Theorem 4. Since $|L| = q + 1$, both choices retain two directions even at $q = 2$. The proof thus justifies this particular pencil deletion; it does not assume that arbitrary deletion preserves connectivity.

Formal scope and attribution

The complete norm-shell proof was developed earlier in this project for the conjectural statement in [VN, Section 4.6]. Its acceptance and the question accounting are prior work; this edition does not count that answer again. Formula (H2), the elementary unitary calculation and the classical root generators are not claimed to be historically new.

The retained module `HermitianPlaneTransport.lean` proves thirteen statements: the Gram-equation equivalence, invariance of the norm under the involution, norm preservation, existence and uniqueness of (H2), and the root-matrix and nonzero-displacement identities. They hold over an arbitrary field with an involution. The retained `ResidueConnectivity.lean` supplies the abstract generating-move connectivity implication. The finite-field identification with $\mathrm{SL}(2, q)$, elementary generation, projective map (H5) and the source geometry applications are written proofs here. They are not formal consequences claimed to have been instantiated in those modules.